SOFTWARE ENGINEERING / RIYADH, SAUDI ARABIA

From concept. To production.

I turn operational problems into well-structured software. From enterprise logistics to adaptable e-commerce, I design, build and own systems people actually rely on.

230K+	30%	02
Consolidated shipments processed through ShipMatrix	Logistics workload automated at Align Technology	Startups adopting the configurable commerce backend

01 / FLAGSHIP CASE STUDY

ShipMatrix

Creator & Backend Engineer | Oct 2023-present

ShipMatrix is a B2B logistics automation platform I designed, built and operated independently. It bridges inconsistent merchant exports and warehouse operations, preparing carrier-ready shipping data without requiring merchants to rebuild their ERP or OMS workflows.

PRODUCTION USE / ALIGN TECHNOLOGY
ShipMatrix supports Align Technology's logistics operations.

What the platform handles

Data interpretation	Fulfillment workflows	Carrier-ready output
Smart Adaptable Mapping learns how different source structures and values relate.	Bulk processing, configurable handling, pick/pack workflows and order consolidation.	Transforms processed shipment data into carrier-specific export formats.

shipmatrix.net

01 / ENGINEERING DEEP DIVE

Smart Adaptable Mapping (SAM)

SAM uses a one-time, 18-question setup to interpret data structures, values and their relationships. When new files arrive, it checks those relationships against the actual upload and adapts the result accordingly.

Traditional template constraints

Each merchant must repeatedly reorganize their source export to match a rigid shipping template.

SAM approach

One saved mapping interprets varied source files without requiring changes to the merchant's existing systems.

Two interpretation stages

01 / One-time setup

Answer mapping questions. Interpret data structure, values and relationships, then save the configured model.

02 / Every new upload

Reinterpret the incoming file; validate value relationships and adapt to actual columns or missing quantities.

Three examples of structural differences

Split address: multiple address columns can be interpreted as a single delivery address.

Repeated items: rows representing the same order/item can be consolidated under configured quantity rules.

Address reference: a configured authoritative source can determine the correct delivery address.

Recorded operational scale
231,719 consolidated shipments processed, representing approximately **1.39 million underlying orders**, based on an actual historical average of **6 orders per consolidated shipment**.

Business impact

This way, not a single change the merchant have to do in their systems to automate their shipments.

02 / REUSABLE BACKEND ENGINEERING

One architecture. Different businesses.

I built a configurable commerce backend on Django, Django Oscar and PostgreSQL, designed for businesses with different products, pricing requirements and fulfillment models. Its reusable foundations can be configured rather than rewritten for each industry.

Reusable configuration

Option groups, values, levels and conditional pricing rules driven by database configuration.

Server-authoritative pricing

The backend computes adjustments and validates totals instead of trusting browser-submitted prices.

Operational architecture

Order workflows, localized Arabic/English behavior and integrations designed for real operational use.

Django pricing-engine demo

A public, independently runnable extract of the pricing logic demonstrates jewelry, fashion and food configurations. It is **not** the private commerce platform, and contains no production data or credentials.

The public repository includes **19 Django tests**, GitHub Actions checks and a downloadable tested demo artifact.

github.com/MOHD97v/django-pricing-engine-demo

YAGARZ

YAGARZ is a jewelry commerce application adopting the configurable backend foundation. Its presentation and customer experience are distinct from the publicly shared pricing-engine demo.

Engineering principle: Build a reliable foundation once, then configure it for different businesses and operational constraints.

03 / EXPERIENCE

Operational insight. Engineering ownership.

Creator & Backend Engineer - ShipMatrix Solutions | Oct 2023-present

Designed, developed and operated logistics automation infrastructure independently, processing more than 230,000 consolidated shipments.

Logistics Supervisor - Align Technology | Mar 2024-Dec 2025

Automated approximately 30% of logistics workload across Saudi Arabia and the UAE using spreadsheet and workflow automations alongside enterprise operational responsibilities.

WMS / Systems Operations - Abudawood Logistics | Feb 2021-May 2023

Warehouse-management systems and operational execution informed a pragmatic approach to automation and system reliability.

04 / ENGINEERING MINDSET

Built for what happens after launch.

Understand the real workflow first. Keep business logic authoritative. Build reusable capabilities. Own the full lifecycle from requirements through maintenance and support.

Tools and disciplines

Python / JavaScript / Django / Django Oscar / PostgreSQL / MySQL / SQLite / GitHub Actions / CI/CD / Automated tests / Rule-driven pricing / Excel and CSV automation / ERP and WMS workflows / Cloudflare Pages and D1

05 / GET IN TOUCH

Have a difficult backend problem?

Riyadh, Saudi Arabia | Backend software engineering and operational automation

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